

Global Constraints Beyond Finite Modal Data

A Closure Criterion for Whole-History Admissibility

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Abstract: Can all finite descriptions of a history determine whether that history is physically admissible? This question matters for theories whose laws constrain complete histories: a candidate may pass every finite test yet violate a condition on the whole history. We establish an exact criterion for when finite modal data suffice. Under the stated separation and extendibility assumptions, admissible histories form a dense subset of the space of coherent complete descriptions. Finite data determine admissibility exactly when that subset is closed; equivalently, every excluded history must have a finite witness to its exclusion. An explicit binary example shows the failure: two history spaces have the same finite descriptions and the same actual history, but admit different complete histories. We then extend the criterion to concurrent configurations, identifying alternative orders of independent events that describe the same completed configuration. The result identifies the global information a faithful reformulation must preserve. It applies to Arborescent Eternalism, an enrichment of the tenseless block with an objective structure of admissible alternatives, while leaving the interpretation of that modal structure open.

Keywords: laws of nature; global constraints; modality; inverse limits; safety and liveness

1 Introduction

When laws constrain complete histories, how much of their content can be recovered from finite descriptions? A history may satisfy every finite test while violating a requirement that concerns its entire course. The question arises in time-symmetric and directionless approaches, where laws delimit physically possible histories without generating later states from earlier ones (Adlam, 2022; Chen & Goldstein, 2022). This paper identifies exactly when finite modal data determine whole-history admissibility.

Finite restrictions determine an inverse limit X_∞ of coherent descriptions. The hierarchy must be adapted: a level- n record may depend only on the portion of a history represented at level n , not on facts available only from the completed future history. Relative to such a hierarchy, a law may nevertheless select only $R \subseteq X_\infty$. Exactly when R is closed, global admissibility can be characterised by exclusions witnessed at finite stages. If R is dense and non-closed, every finite description remains compatible with an admissible history although some coherent complete histories are globally excluded.

The criterion uses the familiar distinction between safety and liveness. A safety requirement has a finite witness whenever it is violated. A liveness requirement can remain compatible with every finite prefix even when a complete history violates it. In the topology used here, safety properties are closed and liveness properties are dense. We use this distinction to diagnose the admissibility information lost when a theory of complete histories is reduced to finite descriptions.

Independent events add a second difficulty: several orders of listing the events can describe the same physical configuration. We separate this redundancy from global admissibility and extend the criterion to completed configurations. An explicit example

combines independent events, compatible reconvergence and a global exclusion invisible at every finite stage.

Arborescent Eternalism (AE) provides an application (Mury, 2026a). AE enriches a tenseless block with an objective organisation of physically admissible alternatives. The results identify which information this organisation must retain when it represents whole-history laws. They leave open whether that information should be interpreted instrumentally, as part of the laws, or as objective modal structure.

Sections 2–4 define the representational task and prove the closure criterion. Sections 5–6 show how finite agreement relations and admissible histories can be reconstructed from a prefix tree with designated paths. Section 7 extends the analysis to concurrent configurations. Sections 8–10 explain what faithful reformulations must preserve and illustrate the criterion. Sections 11–13 discuss objections, the scope of the results and their implications.

2 Whole-history laws and modal organisation

2.1 Complete histories and global constraints

A whole-history formulation begins with a class $\mathcal{H}$ of complete histories and a rule selecting which are physically admissible. Such a rule may be instrumental, nomological or ontic. The structural results below are neutral among these readings.

For the later application, AE uses an event organisation E , a projection Π to a tenseless physical base M , and a distinguished realised history H^* . Weights, global evaluation and agency belong to the wider programme but play no role in the propositions below.

In that application, 'arborescent' is theory-specific rather than graph-theoretically strict. It denotes a rooted organisation with a globally consistent antecedence-to-extension direction; compatible alternatives may branch and reconverge. A merger of incompatible pasts instead loses provenance and requires an enriched representation. Write $Q\Pi$ for the projected configuration graph used in Section 7.

At the discrete level of Section 7, $Q\Pi$ is a rooted configuration graph and $\leq\Pi$ is its reflexive-transitive extension order. Branching and compatible reconvergence are permitted. The prefix tree of Sections 3–6 records a chosen adapted hierarchy of finite descriptions and must not be identified with $Q\Pi$.

The objective interpretation is AE's proposal; the reconstruction theorems remain neutral about it. The mathematical question is which finite and global relations are recoverable from which reduced descriptions. This paper does not use the structural results to infer ontology, temporal orientation, weights or agency.

2.2 Minimal eternalism

Let B_0 denote a minimal description of a complete actual physical history with its tenseless physical relations. This is intentionally thinner than a physical theory equipped with

laws, an entire solution space and modal relations. The physical base M introduced above is not interchangeable with the full field content of that history. The comparison isolates what an actual-history description fixes from what a theory adds about admissible alternatives.

This thinness should not be turned into a criticism by stipulation. A defender of B_0 may simply decline the modal task considered here. If the aim is only to represent the actual four-dimensional history, no branching structure is required. The present argument begins only when a theory is also asked to represent how complete admissible histories share finite records and diverge.

The status of time in quantum gravity bears on, but does not by itself settle, the debate over temporal existence. Le Bihan (2020), for example, argues that the non-fundamentality of time in different quantum-gravity programmes can support different eternalist interpretations, including standard and atemporal eternalism. Likewise, adding a solution space to a tenseless theory need not introduce temporal becoming. A complete history can be one member of a structured family without being generated successively.

2.3 *Three readings of alternatives*

The same family of histories admits at least three broad interpretations.

First, an **instrumental reading** treats the histories as models used for calculation and counterfactual reasoning. Only one history represents reality.

Second, a nomological reading treats the structured family as part of the law: the laws determine not only what actually occurs but the range and relations of admissible complete histories. This reading must still distinguish a merely kinematically allowed model from a genuine physical possibility. Le Bihan et al. (2025) argue that genuine physical possibilities should be associated with the dynamically constrained solution space rather than with unconstrained kinematical models. The present theorem does not settle that interpretive criterion; it presupposes that the members of $\mathcal{H}$ have already passed the relevant admissibility test.

Third, an **ontic reading** takes at least some modal structure to correspond to objective features of reality. This does not yet imply that every history is a concrete world. Objective modality may be grounded in laws, powers, constraints, or global structure.

The results hold under all three readings. This permits the mathematical preservation question to be assessed before choosing an interpretation of possibility. No plurality of concrete worlds is assumed.

2.4 *What reduction to finite modal data would require*

The relevant reduction question is functional: can a description retain the same admissible histories and finite-agreement relations while discarding explicit whole-history admissibility data?

A faithful reformulation may use different mathematical language while retaining the same target relations. Example 1 shows how discarding whole-history admissibility can lose information despite preserving every finite restriction. Sections 6 and 7 specify what must be recoverable in representations of filtered histories and projected configurations.

3 Histories and finite records

Let $\mathcal{H}$ be a non-empty set of complete admissible histories. A history is a complete physical configuration, not a world successively produced in time. Fix a countable hierarchy of finite recorded descriptions. A level records specified information, not necessarily the first n physical events or a simultaneous spacetime slice. A concurrent configuration need not have a unique first n events.

For each finite resolution level $n \in \mathbb{N}$, let

$$r_n: \mathcal{H} \rightarrow X_n$$

assign to a history its finite recorded restriction at level n . Two histories are equivalent at level n when they have the same restriction:

$$h \sim_n h' \quad \text{iff} \quad r_n(h) = r_n(h').$$

A0: Realised restriction spaces. Define $X_n = r_n(\mathcal{H})$, so every r_n is surjective. Equivalently, X_n is identified with $\mathcal{H}/\sim_n$. The word recorded is neutral between an event prefix, a bounded observational algebra, a finite causal record or another physically justified description. The information may be relational even when spacetime is emergent. What matters for the next results is the fixed hierarchy, not a claim that this hierarchy is a preferred physical foliation.

A0': Adapted finite descriptions. For each n , $r_n(h)$ depends only on the portion of h represented at level n . In particular, a finite record may not encode a fact whose determination requires access to the completed future history. This excludes anticipatory codings, such as attaching to every finite prefix the last non-zero index of the entire completed sequence.

Within this convention, impose the following conditions.

A1: Common formal root. $X_0 = \{*\}$.

All histories agree at the coarsest level. A trivial root can be adjoined to a presentation, so this condition is a normalisation, not a proof of a common cosmological source, a low-entropy boundary or physical overlap between worlds. Any stronger physical interpretation of the root needs a separate argument.

A2: Projective refinement. For every n , there is a map

$$p_{n+1,n}: X_{n+1} \rightarrow X_n$$

such that $r_n = p_{n+1,n} \circ r_{n+1}$.

Finer records determine unique coarser records. The assumption gives a unique ancestry of descriptions; it does not forbid several finer continuations of one coarse record.

A3: Finite separation. If $h \neq h'$, then $r_n(h) \neq r_n(h')$ for some finite n .

Distinct histories are distinguishable at some finite level. Without this assumption, two histories may be formally different while agreeing at every finite resolution relevant to the representation.

A4: Nontrivial alternatives. For some n and $x \in X_n$, the set $p_{n+1,n}^{-1}(x)$ contains at least two elements.

Under A0–A3, A4 is equivalent to $\text{card}(\mathcal{H}) \geq 2$. Indeed, two distinct histories differ at a least level $m > 0$, while agreeing at $m - 1$; their restrictions therefore give two children of the same vertex. Conversely, two realised children belong to distinct histories. Thus A4 ensures non-linearity of this encoding, not physical indeterminism. A deterministic theory with varying initial conditions can satisfy it. Moving the split to a later level does not cure this issue when the shared record remains physically incomplete. The least-level argument uses A1 to guarantee the common level-0 root.

These conditions specify a filtered representational task. Whether disagreement represents genuinely distinct physical possibilities, rather than gauge copies or ignorance of an omitted state variable, is an application-dependent question. The distinction between branching and correspondence definitions of determinism is therefore relevant (Müller & Placek, 2018), but A4 is not offered as a new definition of indeterminism. No probability, actualisation rule or concrete plurality of worlds is assumed.

4 From finite modal data to global histories

The local finite-extension data are obtained as follows. For $x \in X_n$ and $k \geq 1$, define

$$F_{n,x}^{(k)} = \{y \in X_{n+k} : p_{n+k,n}(y) = x\},$$

where $p_{n+k,n}$ is the composite bonding map. The family $\mathcal{F}_{n,x} = (F_{n,x}^{(k)})_{k \geq 1}$ includes its induced restriction maps. By contrast, $C_{n,x} = r_n^{-1}(x) \subseteq \mathcal{H}$ is a cylinder of complete histories. Both constructions are induced from $\mathcal{H}$; they are not independently postulated local and global worlds. In fact, A0 and projectivity give $F_{n,x}^{(k)} = r_{n+k}(C_{n,x})$. Finite-extension fibres are finite-level images of cylinders. The relevant difference is that passing to these images can lose information about complete admissibility.

Define the inverse limit

$$X_\infty = \lim_{\leftarrow n} (X_n, p_{n+1,n}) = \{(x_n)_{n \in \mathbb{N}} : p_{n+1,n}(x_{n+1}) = x_n \text{ for every } n\}.$$

Every complete history determines a coherent global ray,

$$\rho : \mathcal{H} \rightarrow X_\infty, \quad \rho(h) = (r_0(h), r_1(h), \dots).$$

A3 makes ρ injective. It does not make it surjective. The distinction is substantive. The finite restriction spaces can admit a coherent tower whose every component is realised at its own level, although no single admissible history realises the whole tower.

Example 1 (a coherent but inadmissible global ray). Let $\mathcal{H}_{\text{fin}}$ consist of infinite binary sequences with finite support, and let r_n retain the first n bits. Then $X_n = \{0,1\}^n$, with ordinary truncation. Every finite binary word extends by zeros to an admissible history, so A0–A4 hold and X_∞ is the full binary sequence space. But $(1,11,111, \dots)$ is not induced by a member of $\mathcal{H}_{\text{fin}}$. Admissibility requires the complete sequence to be eventually zero, a condition that no finite prefix can refute. The ordinary prefix hierarchy is adapted in the sense of A0'.

Compare $\mathcal{H}_{\text{all}} = \{0,1\}^\mathbb{N}$ with $\mathcal{H}_{\text{fin}}$ under the same finite-prefix maps. Both history spaces induce the same $X_n = \{0,1\}^n$ and the same bonding maps; both contain $H^* = (0,0,0,\dots)$ which can be designated as actual. Yet $(1,1,1,\dots)$ is admissible in $\mathcal{H}_{\text{all}}$ and not in $\mathcal{H}_{\text{fin}}$. Thus the actual history and the entire finite restriction system do not, in general, determine global admissibility. Moreover, $\mathcal{H}_{\text{fin}}$ is countable whereas $\mathcal{H}_{\text{all}}$ is uncountable.

Write

$$R = \rho(\mathcal{H}) \subseteq X_\infty$$

for the admissible rays. The pair consisting of the inverse system and R represents the complete filtered modal data. R is determined by $\mathcal{H}$, but it need not be recoverable from the inverse system alone. It is additional data relative to that reduced description, not an independent variable once $\mathcal{H}$ and its restrictions have been fixed.

A5: Local determination of the global. Every coherent family in X_∞ is realised by a history in $\mathcal{H}$.

Proposition 1 (local determination criterion). Under A2, ρ maps $\mathcal{H}$ into X_∞ . Under A3 it is injective. Under A5 it is surjective, so $R = X_\infty$. Under A2, A3 and A5,

$$\mathcal{H} \cong X_\infty.$$

Here the isomorphism is the canonical bijection induced by ρ , preserving every finite restriction coordinate; with the initial topology generated by the r_n , it is a homeomorphism.

The map separates two questions: do the finite descriptions identify a history, and does every coherent sequence of descriptions correspond to an admissible history? Separation answers the first; A5 answers the second.

Proof. A2 gives $p_{n+1,n}(r_{n+1}(h)) = r_n(h)$, so $\rho(h)$ is coherent. A3 gives injectivity. A5 is exactly surjectivity. ■

Thus coherence at every finite level entails whole-history admissibility only under A5.

A5 therefore separates two regimes. When it holds, the inverse system suffices to recover the admissible history space up to its finite records. When it fails, $R \subsetneq X_\infty$ carries a whole-history restriction lost by the finite reduction. Example 1 shows that A5 is not implied

by A0–A4; $\mathcal{H}_{\text{all}}$ shows that it is compatible with them. Neither regime is imposed by eternalism alone.

Give each X_n the discrete topology and X_∞ its subspace topology from the product. A non-empty basic neighbourhood constrains finitely many coordinates; coherence reduces those constraints to one sufficiently fine coordinate. By A0, some admissible history realises that coordinate, so the neighbourhood meets R . Hence $\overline{R} = X_\infty$, and A5 holds exactly when R is closed. Finiteness of every X_n is needed for the usual compactness conclusion, but not for this density-and-closure argument.

Proposition 1.1 (finite-prohibition criterion). In the cylinder topology on X_∞ , R is closed if and only if there exists a family of forbidden finite restrictions such that a coherent complete description belongs to R exactly when none of its finite restrictions is forbidden.

In this criterion, each excluded complete description must already fail at some finite stage.

Proof. If R is closed, $X_\infty \setminus R$ is open and hence a union of basic cylinders; take their generating finite restrictions as forbidden. Conversely, the complement of the histories avoiding a family of forbidden restrictions is a union of cylinders, hence open. $\square$

A closed admissible set therefore has a finite witness for every exclusion. A non-closed set can exclude a complete history that passes every finite test.

The criterion is relative to the chosen finite-description hierarchy. Under A0', a genuinely finite-window dynamical constraint gives a closed admissible set whenever violation is witnessed on some bounded record. By contrast, an all-at-once law in the sense explored by Adlam (2022) or a law-as-constraint framework of the kind discussed by Chen and Goldstein (2022) can impose a whole-history condition whose violation has no finite witness, yielding a dense non-closed R . Without A0', one can trivially encode future global information into a finite record and make the same admissibility condition closed.

The condition is relative to the chosen records: it asks whether their finite tests can detect every global violation. Example 1 exhibits a violation that all such tests miss.

For $U_{n,x} = \{z \in X_\infty : z_n = x\}$, coherence also gives the canonical correspondence

$$\lim_{\leftarrow k} F_{n,x}^{(k)} \cong U_{n,x}, \quad \rho(C_{n,x}) = R \cap U_{n,x}.$$

A coherent finite-extension family determines its coarser coordinates uniquely, and conversely a ray through x determines that family. The same neighbourhood argument makes $R \cap U_{n,x}$ dense in $U_{n,x}$. Thus the fibres determine the coherent continuations through a record; the intersection with R specifies which are globally admissible.

Alpern and Schneider (1985) give the underlying safety/liveness characterisation. The admissible image here is dense because every retained finite record occurs in an admitted history. If that image is also closed, it exhausts the coherent completion, exactly as A5 requires.

Our contribution is to apply this criterion to information lost under finite modal reduction and, in Section 7, to the projected configuration structure used by AE.

The global content is therefore neither gauge holonomy nor a mysterious property of the word fibre. It is the admissible subset of a coherent completion. The construction concerns an inverse limit along a hierarchy, not sheaf gluing on an open cover. In a complete-history formulation R is derived; in the reduced inverse-system formulation it must be supplied unless A5 holds.

5 The canonical prefix structure

Define the graded object

$$T(\mathcal{H}, r) = \coprod_{n \in \mathbb{N}} X_n.$$

Its root is $*$ $\in X_0$. Draw an edge from $x \in X_n$ to each $y \in X_{n+1}$ with $p_{n+1,n}(y) = x$. Each complete history determines a ray

$$\rho(h) = (r_0(h), r_1(h), r_2(h), \dots).$$

Proposition 2 (prefix arborescence). Under A1 and A2, $T(\mathcal{H}, r)$ is a graded rooted tree: every non-root vertex has one immediate parent and a unique directed path from the root. Under A4 it is non-linear.

Each finite description has one immediate coarser restriction. This is what gives the prefix construction its tree shape.

Proof. A2 gives each level- $(n + 1)$ vertex one parent. Iteration reaches the unique vertex of X_0 along one path. Edges strictly increase the grade, so there are no directed cycles or alternative ancestry paths. A4 supplies a vertex with at least two children. $\square$

The resulting tree encodes restrictions; its existence alone leaves the physical interpretation of branching open.

The tree represents agreement at the chosen levels. It is not thereby the event graph E , the projected physical organisation $Q\Pi$ or an ontological overlap of worlds. Belnap (1992) gives an axiomatic branching-space-time framework; Placek and Belnap (2012) construct branching models from solution sets in response to pruning worries. The present prefix construction is narrower and makes its dependence on chosen agreement data explicit.

The tree and its designated admissible rays retain the filtered history information. The tree alone can lose whole-history constraints. The next section states the reconstruction precisely.

This distinction matters physically: a diagram of possible finite extensions can admit more infinite paths than the law permits as complete histories.

6 Reconstruction of filtered histories and graph presentations

There are two different reconstruction statements. The first starts from histories and their agreement relations, without assuming maps from a competing graph. The second makes explicit what a graph presentation must carry to represent those same data. Both are relative to the chosen filtration.

Proposition 3(a) (filtered-history reconstruction). Under A0–A3, the filtered history space $(\mathcal{H}, (\sim_n)_n)$ and the pair (T, R) determine each other up to a bijection preserving every finite agreement relation. The admissible rays cover every vertex of T . Conversely, any graded rooted tree equipped with a set R of infinite rooted rays covering every vertex reconstructs such a filtered history space. A distinguished actual history corresponds to a distinguished ray.

The next result specifies the information needed to recover histories together with all their finite agreement relations.

Proof of (a). Starting from histories, set $X_n = \mathcal{H} / \sim_n$ and use refinement to obtain the parent maps. A3 makes ρ a bijection from $\mathcal{H}$ onto R , and two histories agree at level n exactly when their rays have the same level- n vertex. Conversely, take $\mathcal{H}' = R$ and define agreement by equality at each level. Vertex coverage makes every quotient equal to the original level set. Reconstructing the tree and its admitted rays returns (T, R) . The constructions preserve a designated H^* whenever one is included. $\square$

The reconstruction preserves the stated agreement relations and admissible rays. Ontological uniqueness would require a further argument.

Thus a modal B-theory that supplies a solution set and the nested equivalence relations of finite agreement supplies the inputs for this reconstruction. A bare set of solutions without those relations does not. This is mutual reconstruction of structures, not a claim of formal definitional or Morita equivalence between axiomatic theories: the latter requires specifying languages and interpretations of the kind studied by Barrett and Halvorson (2016).

For (b), let $G = \coprod_n G_n$ be a simple graded directed graph with one root in G_0 , with edges only from level n to level $n + 1$. A **prefix-adequate presentation** consists of maps $q_n: G_n \rightarrow X_n$ and rooted infinite paths $\eta(h)$ for all $h \in \mathcal{H}$, such that:

- (i) $q_n(\eta(h)_n) = r_n(h)$ at every level;
- (ii) every edge $g \rightarrow g'$ satisfies $q_n(g) = p_{n+1,n}(q_{n+1}(g'))$;
- (iii) every vertex and edge lies on at least one represented path $\eta(h)$.

Set $g \equiv g'$ when they have the same grade and the same q_n -image. The quotient is taken as a simple graph, with multiple copies of an edge identified.

Proposition 3(b). A prefix-adequate presentation admits a surjective grade-preserving graph morphism $q: G \rightarrow T$, with kernel $\equiv$, and $G/\equiv \cong T$. Its induced map on infinite paths, denoted q_∞ , satisfies

$$q_\infty \circ \eta = \rho, \quad q_\infty(\eta(\mathcal{H})) = R.$$

A more detailed graph can represent the same filtered histories if its extra distinctions disappear under the specified quotient.

Proof of (b). Define $q(g) = q_n(g)$. Condition (ii) makes it a graph morphism. For each $x \in X_n$, A0 supplies h with $r_n(h) = x$, and condition (i) supplies a preimage vertex. Every edge of T is likewise lifted along a history realising its finer endpoint. Equality of the images is precisely $\equiv$, so the quotient is bijective on vertices and on the simple edge relation. Applying q_n at every coordinate of $\eta(h)$ proves the path identity and its image statement. $\square$

The quotient therefore removes graph distinctions that leave the finite-history structure unchanged.

Condition (iii) excludes unused presentation surplus; it is not the reason for surjectivity. No parent map on G was assumed. If all q_n are injective, the graph presentation itself is isomorphic to T . In general, the quotient, not a literal subgraph of G , recovers the tree.

The global content is carried by the designated represented paths, not by every path that the bare graph allows. Call a presentation with admissible path set $\mathcal{R}_G$ **globally adequate for this task** when $q_\infty(\mathcal{R}_G) = R$; the specified set $\mathcal{R}_G = \eta(\mathcal{H})$ meets this condition. Allowing extra graph paths without checking this equality may change the global modal theory.

Corollary 3.1 (scope of elimination). A representation from which the same histories and finite agreement relations can be recovered also recovers (T, R) . Discarding R can fail to preserve global admissibility even when T and the actual ray are fixed. Example 1 witnesses that failure.

A change of notation preserves the target when the same histories, agreements and exclusions remain recoverable. The preservation requirement concerns those relations, irrespective of whether they are drawn as a tree.

The first clause is a reconstruction requirement, not a surprising argument that faithful encodings preserve their encoded information. The second identifies a substantive loss under a specific reduction. Neither requires every physical representation to be a graph, to use the same labels, or to retain every redundant intermediate vertex.

The mathematical invariant at this stage is the filtered history structure, not a unique fibre formalism or the complete physical projection. Weatherall (2019a, 2019b) provides the broader context of theoretical equivalence; the preservation criterion here is narrower and explicitly stated. The physical relations involving independent events require the separate treatment that follows.

7 Projected configurations, reconvergence and global admissibility

The target in AE is not that E or QII be a strict graph-theoretic arborescence. QII is a rooted directed graph of projected physical configurations. Its reachability relation supplies the

antecedence-to-extension order. Compatible configurations may branch and reconverge. A reconvergence that identifies incompatible pasts is instead a coarse physical state-identification and loses historical provenance; the local-to-global theorem below does not quotient such histories together.

For an independence diamond, use distinct compatible event occurrences a and b that are concurrent extensions of a configuration C . The two serialisations $C \rightarrow C \cup \{a\} \rightarrow C \cup \{a, b\}$ and $C \rightarrow C \cup \{b\} \rightarrow C \cup \{a, b\}$ reach the same configuration. Occurrences are not action labels: repeated uses of one label remain distinct events. This is the paradigm of a coherent reconvergence. If two incompatible histories are mapped to one later coarse physical state, that identification belongs to a different, provenance-losing quotient and must be represented separately.

$$C \rightarrow C \cup \{a\} \rightarrow C \cup \{a, b\}, \quad C \rightarrow C \cup \{b\} \rightarrow C \cup \{a, b\}.$$

Let $T_{\text{path}}(Q\Pi)$ be the standard unfolding whose vertices are finite directed paths from the root, with an edge given by appending one graph edge. The endpoint map ϵ sends each path to its endpoint. This is the familiar unfolding move used in concurrency semantics; it is auxiliary and is not itself the physical target (Nielsen et al., 1981).

$$\epsilon: T_{\text{path}}(Q\Pi) \rightarrow Q\Pi.$$

Proposition 4 (standard endpoint reconstruction). $T_{\text{path}}(Q\Pi)$ is a rooted tree and $\epsilon: T_{\text{path}}(Q\Pi) \rightarrow Q\Pi$ is a surjective graph morphism. Quotienting paths exactly by equality of endpoints recovers $Q\Pi$ as a simple directed graph. The endpoint map is injective exactly when every vertex has a unique directed path from the root. Thus reconvergence is precisely what makes the unfolding contain representational multiplicity.

Independent events can be listed in different orders while reaching the same physical configuration. The endpoint relation records which paths describe that same configuration.

Proof. Every non-empty path has one predecessor, obtained by deleting its final edge, so the path graph is a rooted tree independently of whether $Q\Pi$ itself contains cycles. Reachability gives surjectivity on vertices, and appending any edge to a path reaching its source lifts that edge. Equality of endpoints is exactly the kernel relation of ϵ , so the simple quotient has the same vertices and edges as $Q\Pi$. Finally, ϵ is injective exactly when no vertex has two distinct root paths. $\square$

Quotienting by endpoint identity recovers the configuration graph. Without that relation, the path tree can count alternative descriptions of one configuration separately.

$$T_{\text{path}}(Q\Pi) / \sim_{\epsilon} \cong Q\Pi.$$

A diamond therefore makes ϵ non-injective without making the projected organisation inconsistent. Proposition 4 itself applies to arbitrary graph reconvergence, but Section 7.1 uses only coherent reconvergences when histories are represented by ideals. This is not a counterexample to Proposition 2: A2 gives a parent function between levels of one chosen filtration, whereas $Q\Pi$ represents physical configurations and can have several compatible

immediate antecedents. In AE terminology, 'arborescent' refers to the rooted global extension orientation, not to the strict graph-theoretic property characterised by injectivity of ε .

The unfolding and endpoint relation together retain $Q\Pi$; the tree alone need not. This fact is standard rather than a new AE theorem. Its role here is diagnostic: it identifies which representational multiplicity belongs to serialisation and which endpoint identifications belong to the projected physical model. Nielsen, Plotkin and Winskel (1981) provide the classical concurrency setting in which unfoldings, occurrence structures and event domains are related.

Nor may $\text{Tpath}(Q\Pi)$ be identified automatically with the prefix tree T of Sections 3–6. Such an identification requires compatible serialisations and restrictions. For configurations rather than words, 'take the first n events' need not be single-valued. The local-to-global result therefore has to be restated on $Q\Pi$ itself rather than imported from the auxiliary filtration. The next proposition supplies that bridge.

7.1 *Projected completion and admissibility*

Let $\leq\Pi$ be the reflexive-transitive closure of the extension edges of $Q\Pi$. In the acyclic models considered here, $(Q\Pi, \leq\Pi)$ is a partial order. For Proposition 5, let $Q\Pi^c$ be the projected configuration order of finite compatible configurations: whenever $C \leq\Pi E$ and $D \leq\Pi E$, C and D are jointly co-realizable within the configuration represented by E . This is satisfied by ordinary event-structure configurations ordered by extension and covers independence diamonds and other reconvergences of compatible antecedents. A merger of incompatible pasts is not in $Q\Pi^c$ unless the representation is enriched with provenance.

Let $X\Pi = \text{MaxIdl}(Q\Pi^c)$, the set of maximal ideals of the coherent extension order (including principal ideals of maximal configurations). These are the completed projected possibilities, the analogue of X_∞ . For $C \in Q\Pi^c$ define $U_C = \{X \in X\Pi : C \in X\}$. The cylinders form a basis: if $X \in U_C \cap U_D$, directedness gives $E \in X$ above C and D , and $U_E \subseteq U_C \cap U_D$; conversely the intersection is the union of all such U_E . Let $R\Pi \subseteq X\Pi$ be the maximal ideals induced by physically admissible complete histories. Every ideal is contained in a maximal ideal by Zorn's lemma; in the countable case recursive extension suffices.

$P0\Pi$ (finite realisability). Every $C \in Q\Pi^c$ belongs to at least one $X \in R\Pi$. Thus every finite coherent projected configuration retained in the model occurs within some admissible complete projected history. A finite configuration contained in no admissible history violates $P0\Pi$.

This premise is the projected analogue of defining finite restriction spaces from histories actually admitted by the theory. It is a substantive coverage condition on $Q\Pi^c$, not a consequence of graph reachability alone. Unlike an arbitrary filtration r_n , the finite data here are the finite configurations themselves, so the construction is intrinsically adapted to the extension order.

Proposition 5 (projected local-to-global criterion). Let $X\Pi = \text{MaxIdl}(Q\Pi^c)$ and assume $P0\Pi$. Then $R\Pi$ is dense in $X\Pi$ for the cylinder topology. Projected local determination, $A5\Pi$, is the condition $R\Pi = X\Pi$. It holds if and only if $R\Pi$ is closed in $X\Pi$.

$$A5\Pi : R\Pi = X\Pi.$$

If $Q\Pi^c$ is countable, every $X \in \Pi$ contains a countable cofinal chain and is generated by a complete extension path. Restrict path serialisations to those whose generated maximal ideal lies in Π , and identify two such paths exactly when they generate the same maximal ideal. The quotient is then naturally bijective with Π . Alternative serialisations of compatible concurrent events therefore do not multiply global possibilities.

The projected criterion asks whether every coherent completion of finite compatible configurations is admitted by the law.

Proof. Let U_C be a basic cylinder. By $P0\Pi$ there is $X \in \Pi$ containing C , hence $U_C \cap \Pi \neq \emptyset$. Since the cylinders form a basis, Π is dense. A dense subset is closed exactly when it is the whole space, proving the equivalence with $A5\Pi$. For the countable path statement, enumerate an ideal X and recursively choose an increasing cofinal chain using directedness. Consecutive chain elements can be joined by finite extension paths whose vertices remain in X because X is downward closed. Concatenation gives a complete path generating X . Prefix the chain by a finite path from the root to its first element, using rooted accessibility. Quotienting by equality of the generated maximal ideal gives the claimed bijection. $\square$

Finite realisability makes admissible completions dense. Closure is then exactly the additional condition that makes them exhaust the completed configuration space.

Corollary 5.1 (same finite projected content, different global theory). Let $X\Pi = \text{MaxIdl}(Q\Pi^c)$, assume $P0\Pi$, and suppose Π is a proper dense subset of $X\Pi$. Then every finite coherent projected configuration is realised within an admissible history, while at least one completed projected possibility is globally inadmissible. No test confined to finite cylinder membership can recover the global admissibility rule. The same finite data $(Q\Pi^c, P0\Pi)$ are compatible with $\Pi = X\Pi$, so finite-cylinder data alone cannot distinguish the admissibility rules.

Independent events can form reconvergence diamonds while a whole-history condition still excludes some completions. Removing redundant orders of events does not remove that global exclusion.

Example 2 (concurrency with a proper dense admissible set). For each non-empty binary word w introduce a choice event $c(w)$, ordered by proper-prefix dependence, with $c(w0)$ and $c(w1)$ in conflict. Add an independent chain $d_1 \leq d_2 \leq \dots$, independent of all choice events. Finite configurations consist of a finite binary choice word and a finite d -prefix. Choice extensions commute with d -extensions, producing independence diamonds; hence the configuration graph is not a tree and the path endpoint map is non-injective. Maximal ideals correspond to infinite binary sequences together with the full d -chain. Let Π contain exactly those maximal ideals whose binary sequence is eventually zero. Every finite configuration extends to such an ideal, so $P0\Pi$ holds and Π is dense. It is proper and non-closed: the constant-one sequence lies outside Π and is approached by admissible eventually-zero sequences. Thus serialisation quotienting and global non-closure coexist in a genuinely non-tree configuration order.

The binary finite-support construction of Section 4 is the special tree case. Proposition 5 does not claim a new domain-theoretic theorem: event structures, domains and unfoldings are standard in concurrency theory (Nielsen, Plotkin, & Winskel, 1981), and trace theory gives a complementary treatment of independent serialisations (Diekert & Rozenberg, 1995). The contribution here is to locate AE's local-to-global gap on the maximal-ideal completion of its coherent projected configuration order. This treatment deliberately preserves incompatible pasts rather than collapsing them into one ideal.

An infinite serialisation can be unfair: it may postpone forever an event that remains independently available. Such a path can have every finite prefix realised while failing to generate a maximal completed possibility. This is a path-level liveness failure, familiar from Alpern and Schneider (1985), not a new physical history. Restricting the path quotient to serialisations generating members of $X\Pi$ removes that artefact and leaves the physical question $R\Pi \subseteq X\Pi$.

The three results address distinct layers of the representation. Proposition 3 recovers histories relative to a chosen filtration. Proposition 4 separates configurations from the paths used to list their events. Proposition 5 asks which completed configurations the law admits. The remaining physical question is whether the law admits every coherent completion or imposes a further whole-history constraint. Section 10.4 illustrates the distinction with limiting frequencies. Asymptotic and boundary-at-infinity constraints are candidates for physical applications; no fundamental physical example is established here.

8 Representation, invariant content and surplus

A faithful reformulation may replace trees, fibres or graphs by relational, presheaf or categorical structures provided it preserves the target content. The relevant methodological distinction is between invariant content and representational surplus (Weatherall, 2016): finite restrictions, compatibility maps, endpoint identifications and whole-history admissibility must be preserved when the reformulation undertakes to represent those functions.

9 What survives reduction to finite modal data?

The central invariant is not a particular fibre or tree vocabulary. It is the combination of finite restriction data, their coherence relations and the admissible subset of the coherent completion. A reformulation preserving those data is faithful for the present target; keeping only the coherent completion can lose whole-history admissibility.

Propositions 3–5 separate three preservation questions: recovery of filtered histories and their admitted rays; recovery of a projected configuration graph from path serialisations and endpoint identifications; and recovery of global admissibility on the completed configuration order. These are conditional representational requirements, not claims that one mathematical presentation is ontologically privileged.

10 Global constraints, B-theories and physical illustrations

10.1 A block without an explicit modal law

B_0 specifies one actual history and its physical relations. It may decline to represent other admissible histories. On the singleton history space A4 fails; that says nothing about the solution space of any laws one might additionally supply. The reconstruction theorem does not require a modal extension unless the explanatory task includes it.

This is not a defeat for B_0 . It is a division of labour. A minimal theory is cheaper because it does less. Whether that is a virtue depends on which phenomena and explanations are demanded.

10.2 Boundary conditions and whole-history constraints

A Past-Hypothesis theory adds a special low-entropy boundary condition and an appropriate statistical postulate. This package addresses thermodynamic typicality. It need not represent the same filtered or projected modal organisation as AE. If typicality itself is used to restrict which complete histories count as admissible, however, the distinction between dense finite realisability and global admissibility becomes directly relevant; the limiting-frequency example in Section 10.4 illustrates this possibility.

Nothing here shows that AE explains thermodynamics better than the Past Hypothesis. The present results concern modal reconstruction and global admissibility. Orientation, weighting, physical projection measures and selection require separate arguments and are not premises of the local-to-global theorem.

10.3 A matched modal B-theory

A modal B-theory B_{modal} supplies a space of solutions or histories and specified relations of finite agreement. By Proposition 3(a), it can reconstruct (T, R) directly from those data, without first introducing an auxiliary graph. If it also undertakes to reproduce AE's projected physical organisation, it must specify the relevant projection, dependencies, incompatibilities and physical identifications. These are separate preservation requirements.

AE is itself an enrichment of the B-theoretic block. A formulation in B-theoretic language that retains the same histories, finite agreements and projected relations retains the structures identified here. This leaves three questions about their role and interpretation:

whether the modal structure is explicit or hidden in the model class;

whether it plays a lawlike or explanatory role;

how it should be interpreted ontologically.

The reconstruction result settles none of these interpretive questions. It states a baseline: a B_{modal} retaining the same filtered history space represents the same (T, R) . If it retains the finite restriction system but changes R , it changes global admissibility. If it retains only a path

tree but changes the endpoint identifications, it can change the projected configuration structure. These are distinguishable reductions, not one undifferentiated elimination of fibres.

10.4 Local, global and asymptotic constraints

Consider a finite-state stochastic model with a fixed initial state and a specified transition kernel. Let $\mathcal{H}$ contain all infinite trajectories whose finite prefixes have positive probability; let r_n retain the first n states, with r_0 empty. A0–A3 hold, and a reachable state with two positive outgoing transitions supplies A4. Any coherent family of these prefixes assembles into a trajectory satisfying the same support conditions, so A5 holds. This follows from the definition of the full support trajectory space, not from the measure-extension theorem. Whether these trajectories are genuine physical possibilities remains an interpretive commitment about the dynamical model (Le Bihan et al., 2025).

Example 1 gives a different possibility: an additional whole-history condition, here eventual vanishing, excludes rays without excluding any finite prefix. This provides a toy comparison with all-at-once constraints on histories (Adlam, 2022), not an identification with a particular fundamental law. Likewise, global dependence in event-horizon descriptions motivates taking whole-history structure seriously (Baron & Le Bihan, 2023), but does not itself prove failure of A5.

For a deterministic evolution with one fixed complete initial state and a unique admissible solution, $\mathcal{H}$ is a singleton and A4 fails. With varying initial states, the same deterministic law can have many histories and satisfy A4 after a common formal root is adjoined. Coarser records can also show later splits because they omit state variables. The correct question is what physical data have been held fixed, not whether the prefix diagram branches somewhere. This is why branching and correspondence accounts of determinism require care about physical identification (Müller & Placek, 2018).

A statistical restriction illustrates the same distinction without changing AE’s meaning of possibility. In binary sequence space, the set of histories with limiting frequency one half admits every finite prefix, since each can be continued by an alternating tail, but excludes the constant-one history. Taking that set as admissible would therefore violate A5. Treating typicality as an admissibility law is, however, an additional choice: a probability-one class is not automatically the entire physically possible class. As defined here, AE keeps admissibility, reference weights and actuality distinct.

Physical-type illustration. Whole-history finiteness conditions have the same logical form. Square-summability, $\sum_n x_n^2 < \infty$, is not determined by any finite prefix: every finite real sequence extends by zeros to an ℓ^2 sequence, while coherent complete sequences outside ℓ^2 remain. Thus ℓ^2 is dense and non-closed in the prefix topology. Conditions such as finite total action, finite radiated energy, normalisability or asymptotic decay can play analogous roles in physical models. This illustrates a type of global constraint; it is not a claim that any one is a fundamental law.

A variational selector provides a second, deliberately bounded illustration. Once E , Π and w are fixed, a functional $J: \text{Hist_phys}(E) \rightarrow \mathbb{R}$ can evaluate an entire admissible history, for example $J(H) = -\log w_E(H) + B_{E,\Pi}(H) + C_A(H)$. Its score or minimiser need not be fixed by any one finite prefix when $B_{E,\Pi}$ or C_A depends on completed-history data. The accompanying finite calculation shows only that such a common generator can yield a unique and perturbatively stable minimiser without history-specific fitting. Its script and output are archived at [https://github.com/Legiolab/arborescent_eternalism_public/tree/0b30398bf80aa72fd47a040bf124080c9f4a1695/proofs/paper-c-modal-rarity/computational/j_reference_calibrated_v6_73/\(j_reference_calibrated_gate.py and results.json\)](https://github.com/Legiolab/arborescent_eternalism_public/tree/0b30398bf80aa72fd47a040bf124080c9f4a1695/proofs/paper-c-modal-rarity/computational/j_reference_calibrated_v6_73/(j_reference_calibrated_gate.py%20and%20results.json)). It is not used in Propositions 3–5 and does not prove that every representation of the same physics must retain J as a primitive; a dynamically or locally reformulated theory may encode functionally equivalent information. The structural claim of this paper remains the loss of global admissibility under specified reductions, not a new non-eliminability theorem for variational functionals.

11 Objections

11.1 The assumptions already contain the conclusion

A1 normalises the representation, A2 supplies unique ancestry of recorded descriptions, and A4 ensures non-linearity. Under A0–A3, A4 is equivalent to having more than one history. None of this derives a physical source or indeterminism. A5 instead determines whether every coherent ray is admitted; it is not needed for the tree construction.

The dependencies should therefore be read as a scope analysis, not as independent discoveries of physical structure:

A1. Without a single level-zero class the diagram is a forest. Adjoining a formal root normalises it; a cosmological interpretation does not follow.

A2. Without a unique coarser restriction, the layered incidence diagram can have multiple parents. A physical diamond is not a contradiction: it belongs to the configuration graph of Section 7, not automatically to this hierarchy.

A3. If distinct histories agree at every level, the rays recover only their observational equivalence classes, not the original histories.

A4. With only one history, the canonical diagram is linear. Multiple deterministic solutions can satisfy A4 when the initial state is not fixed.

A5. Finite-support binary histories show that coherent rays can remain globally inadmissible. Restoring those rays changes the complete modal specification.

The physical burden is to justify the agreement relations, the admissible histories and their projection. Part of this choice is representational; part can be constrained by a physical theory. The reconstruction theorem cannot replace that justification. A0 makes explicit which descriptors have actual representatives within the admitted history space.

11.2 The histories are merely mathematical models

Nothing prevents an instrumental interpretation. The theorem remains valid on that interpretation, but its conclusion is correspondingly representational. It would be invalid to infer directly that every history is concrete.

Still, calling the histories models does not make their structure disappear. Models may stand in relations essential to counterfactual, explanatory and lawlike reasoning. The ontological question is whether those relations track objective modality, not whether the mathematics exists.

11.3 A presheaf eliminates the tree

A presheaf, relational or categorical presentation can replace a particular graphical encoding. If it permits recovery of the same histories and finite agreements, it also permits the construction of (T, R) . If it additionally preserves the projected physical relations, the graph reconstruction of Section 7 applies in the stated discrete class. The claim is not that an actual tree must occur as a subobject of every formalism.

Equivalent descriptions must be compared at the level of the stated target. Preserving only the finite tree is weaker than preserving its admissible rays; preserving a path unfolding is weaker than preserving its physical endpoint identifications. A change of mathematical language is not by itself evidence that either target has been lost.

11.4 Branching conflicts with eternalism

Branching would conflict with eternalism if branches were stages successively coming into existence. That is not assumed. $\mathcal{H}$ contains complete histories, and the prefix relations order their finite restrictions. The structure is tenseless. One history may be distinguished as actual without the others being future entities waiting to become real.

11.5 The theorem is empirically idle

The results give a representational test, rather than a new empirical prediction. They show that finite data and an actual history can underdetermine global admissibility, and specify what two representational tasks require a reformulation to preserve. Empirical discrimination would need additional physical laws for projection, weights or selection.

11.6 The result is too elementary to be novel

Prefix trees, inverse limits, branch restrictions and the safety/liveness topology predate this proposal. The manuscript claims an application and a clarification of dependencies, not their invention. Its contribution must be assessed through the joint argument:

it fixes the functional target before drawing ontological conclusions;

it reconstructs filtered histories from their prefix tree and designated admissible rays;

it distinguishes coherent finite descriptions from the admissible image of complete histories, and shows what is lost on forgetting that image;

it separates the projected physical configuration graph from its path unfolding and from an arbitrary finite-record hierarchy;

it separates indexed modal content from smooth bundle structure;

it distinguishes a matched reformulation from a reduction that changes either global admissibility or physical identifications.

The proposed contribution is philosophical: the paper isolates a loss of lawlike information that a faithful reformulation must avoid. The counterexample remains substantive even though the reconstruction machinery is standard. A general explanatory preference for AE would require a wider comparison.

11.7 Restricted bundles of histories

A tree with a designated covering set of histories is a bundled-tree presentation; the pair (T, R) is therefore not a new mathematical kind. Branching-time semantics has long debated whether every maximal tree path should count as a history. Thomason (1984, pp. 151–152) and Belnap, Perloff and Xu (2001, pp. 200–203) object that restricting histories by a bundle can yield problematic truth assignments in certain open-future scenarios; Spolaore and Zanardo's Stanford Encyclopedia survey (2025) records both this objection and replies to it. AE does not use R as an Ockhamist semantics for what is presently open. R is an all-at-once admissibility constraint on complete tenseless histories. The objection therefore does not disappear, but it changes form: AE must justify the physical law selecting R rather than infer admissibility from the bare tree.

The relevant distinction in Example 1 is between every finite restriction of a candidate being realised somewhere and one admissible complete history realising all of them together. A5 licenses the stronger conclusion; it must not be imported by treating locally extendable choices as freely concatenable. Proposition 5 makes the same distinction after quotienting path serialisations on $Q\Pi$. A theory of genuinely open local choice may impose A5 or A5 Π and reject the bundled restriction; an all-at-once theory may instead supply an independent whole-history law. The paper's claim is the underdetermination between these regimes, not the novelty of bundled semantics.

12 Four grades of indispensability

Four senses of indispensability separate what the results establish from the further explanatory and ontological claims.

Syntactic indispensability would mean that the vocabulary of trees or fibres cannot be removed. This is false. Equivalent languages are available.

Representational indispensability means recoverability of a specified target under faithful reformulation. Proposition 3 gives this for filtered histories and their admissible rays. Proposition 4 addresses the projected graph and its endpoint identifications. Proposition 5 adds the global admissibility datum on the coherent projected completion: finite realisability alone need not determine RII. These are preservation results for stated targets, not a proof that AE is the unique ontology. The corresponding whole-history admissibility datum is R .

Explanatory or nomological indispensability would mean that the retained structure is required by the best explanation or law. The reconstruction theorems do not establish that ranking. They specify what competing descriptions must account for if they undertake the same modal and physical tasks. Their comparative economy and explanatory success remain questions for the wider AE programme.

Ontological indispensability means that reality itself contains objective modal structure corresponding to the invariant. The theorem alone does not establish this. An ontic interpretation needs an additional argument connecting successful representation and explanation to ontology.

An actualist can accept these reconstruction results while denying that non-actual histories are concrete. Likewise, equality of recorded descriptions does not settle whether histories literally share events or instead contain merely corresponding events, a distinction related to Lewis's branching-versus-divergence discussion (Lewis, 1986, ch. 4). The coherent-ideal construction used in Proposition 5 does not identify incompatible pasts; any physical merger that does so requires an explicitly provenance-sensitive representation. AE's objective interpretation needs its own argument; the quotient construction does not supply it.

13 Conclusion

A complete actual history need not determine its alternatives, even when all finite compatibility data are included. The paired history spaces of Example 1 establish that claim explicitly. For a chosen separating filtration, the complete modal specification is mutually recoverable with a prefix tree and its admitted rays. Forgetting the admitted-ray set may change the theory; A5 states exactly when no such information is lost.

The projected physical organisation must be kept distinct from the auxiliary prefix encoding. AE uses 'arborescent' for a rooted, globally directed extension organisation that can branch and reconverge; it does not mean a strict graph-theoretic arborescence. Standard path unfolding separates serialisation multiplicity from physical endpoint identity. Proposition 5 then carries the local-to-global question to the quotient configuration level: finite projected realisability can coexist with a proper dense set of globally admissible completed configurations.

The paper identifies whole-history admissibility as information that can be lost under finite reduction, including after redundant serialisations of independent events have been removed. The closure criterion locates exactly when this loss occurs. Any reformulation undertaking the same task must preserve the relevant global datum. AE gives that structure an

objective explanatory interpretation; the mathematical results establish the preservation requirement while leaving that interpretation open.

Declaration on the use of generative AI

Generative AI systems, including OpenAI's ChatGPT and Codex and Anthropic's Claude, were used during the development of this manuscript for conceptual exploration, adversarial review, formalisation, drafting and structural revision, and to assist with checking proof structure, examples and bibliographic details. The theorem statements, choice of assumptions, interpretation of the results and final editorial decisions are those of the author. No AI system is credited as an author or treated as an evidential source.

All AI-assisted material retained in the manuscript was checked, revised and approved by the author, who accepts full responsibility for the arguments, calculations, citations and final text. AI-assisted work was organised through a research harness tracking assumptions, experiments, evidence and decisions. Numerical results were produced by executable Python scripts, and arguments underwent adversarial review; responsibility for the submitted claims remains with the author.

Proof and source availability

A self-contained statement and proof of the prefix-factorisation criterion, together with supporting derivations and reproducibility material, are available in the online proof repository:
https://github.com/Legiolab/arborescent_eternalism_public/tree/main/proofs/paper-b-global-constraints/.

The repository uses a compressed presentation: assumptions A1–A4 match those of this article, A0 is incorporated into the definition of the realised restriction spaces, and A0' and A5 are not required for the factorisation theorem. Its README provides the correspondence between proposition numbers. The additional local-to-global closure results are proved in this article.

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